Sentential negation causes illusory licensing of both NPIs and NCIs in Czech

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NPI ILLUSIONS

- Unlicenced **negative polarity items** (NPIs) give rise to **grammaticality illusions** only when a certain type of intervening illusory licensor is present:

1) *The student who **no** teacher likes has **ever** got an A.
2) *The student who the teacher **doesn't** like has **ever** got an A.

- NPI illusions are **highly selective** when it comes to the **type of licensor:**
  - Illusions occur with **negative quantifiers** (1) but not with **sentential negation** (2)
- This is argued by psycholinguists to be consistent with the **active scope, scalar theories** and **feature percolation accounts** but not the **cue-based retrieval theory**

Drenhaus et al. (2005), Vasishth et al. (2008),Parker & Philips (2016); Orth et al. (2021), Muller (2022),Schwab (2022), Eberhard et al. (2005)

NEGATIVE CONCORD (NCI) ILLUSIONS

- Languages like Czech exhibit **negative concord**, a requirement for any **negative concord item** (NCI) in a clause to be accompanied by sentential negation but yielding only one logical negation:

3) Jan **ne-viděl** na zahradě **žádného** psa.
   Jan NEG-saw on garden no dog.
   'Jan didn't see any dog in the garden. / Jan saw no dog in the garden.'

- An influential theory claims that **negative concord is syntactic agreement** and as such differs from NPI licensing which is semantic in nature
- Recently, Lacina (2025) found evidence of **NCI illusions** with negative quantifers as illusory licensors/agreement targets in Czech
- If NCI licensing is in the syntax, it should behave similarly to **agreement attraction** (where cue-based retrievel has been successful) as only the negative marker *ne-* is required

→ **Sentential negation** should give rise to **illusions with NCIs** but **not NPIs**

Zeijlstra (2004, 2008; 2022), Lacina (2025), Wagers et al. (2009)

DOES SENTENTIAL NEGATION IN CZECH GIVE RISE TO NCI BUT NOT NPI ILLUSIONS?

WHAT DOES THIS MEAN?

- Illusions with NCI were predicted (syntactic agreement), but the effect with NPIs was unexpected, according to active scope and scalar theories, sentential negation should not be an illusory licensor
  - This casts **doubt on the generalisability of the active scope and scalar theories** cross-linguistically
- The results are **compatible with cue-based retrieval theories** as well as **feature percolation accounts**
- Feature percolation could explain the cross-linguistic difference with English by **appealing to the placement of negation** in the syntactic structure
- While in English, sentential negation is low (weaker percolation), in Czech the negative operator sits high in the structure (strong percolation)

EXPERIMENTS

- Exp 1 (N = 143) tested illusions with the **strong NPI** *ani jeden* ('even one')

  → No illusion predicted ✗

- Exp 2 (N = 179) tested the same sentences with the **NCI** *žádný* ('no')

  → Grammaticality illusions predicted ✓

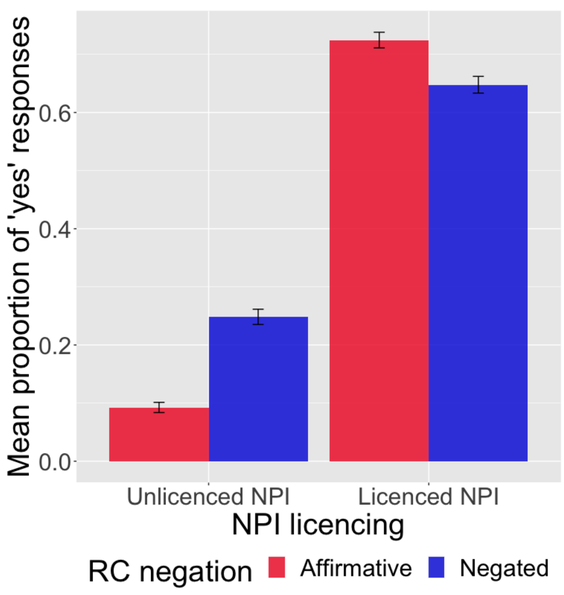
CONDITION	SENTENCE (ITEMS = 32, RSVP AT 350MS PER WORD)
Licensed & No RC Negation	Minulý měsíc ne -vyhrál karbaník, který hledá dlouhodobou práci, ani jednu/žádnou sázku. Last month NEG-won. gambler. who seeks long-term. job even one /no bet
Licensed & RC Negation	Minulý měsíc ne -vyhrál karbaník, který ne. -hledá dlouhodobou práci, ani jednu/žádnou sázku. Last month NEG-won gambler. who NEG-seeks long-term. job even one /no bet
Unlicensed & No RC Negation	*Minulý měsíc vyhrál karbaník, který hledá dlouhodobou práci, ani jednu/žádnou sázku. Last month won gambler who seeks long-term job even one /no bet
Unlicensed & RC Negation	*Minulý měsíc vyhrál karbaník, který ne -hledá dlouhodobou práci, **ani jednu/žádnou** sázku. Last month won gambler who NEG-seeks long-term job **even one /no** bet
'Last month, the gambler, who is/isn't looking for a long-term job, won/didn't win even a single/no bet. '	

- Exp 3 (N = 70) tested sentences with intransitive verbs in the RC with unlicensed **NPIs** in the matrix clause to rule out a parsing explanation

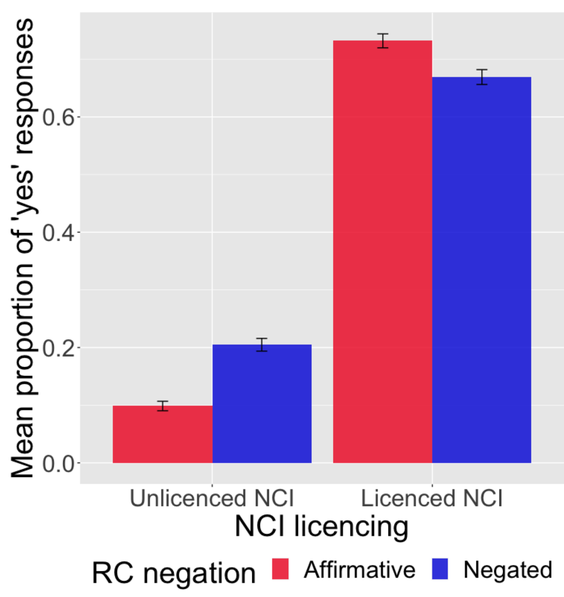
CRITICAL COND.	SENTENCE (ITEMS = 32, RSVP AT 350MS PER WORD)
Unlicensed & RC Negation	*Minulý měsíc vyhrál karbaník, který **ne -přespává v hotelu**, ani jednu sázku. Last month won gambler who **NEG-sleeps.over in hotel** even one bet
'Last month, the gambler, who is/isn't sleeping over in a hotel, won/didn't win even a single/no bet. '	

RESULTS

EXP 1

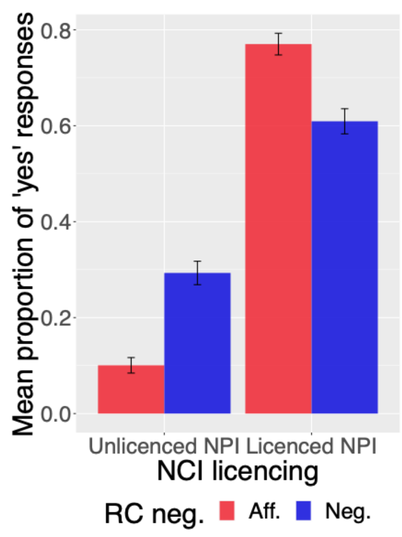


EXP 2



NESTED EFFECT (EXP 1)	BETA	ST.E.	Z-VALUE	P-VALUE
Intercept	-0.63466	0.11945	-5.313	<0.001***
Licensing	1.80260	0.09272	19.442	<0.001***
Licenced : RC Negation	-0.18633	0.06531	-2.853	0.00433**
Unlicenced : RC Negation	0.65349	0.08086	8.082	**<0.001*** **

NESTED EFFECT (EXP 2)	BETA	ST.E.	Z-VALUE	P-VALUE
Intercept	-0.55421	0.11353	-4.882	<0.001***
Licensing	1.57511	0.08368	18.822	<0.001***
Licenced : RC Negation	-0.20701	0.05894	-3.512	<0.001***
Unlicenced : RC Negation	0.78989	0.07869	10.038	**<0.001*** **



EXP 3

EXP 3	BETA	ST.E.	Z	P
Intercept	-2.6093	0.2573	-10.14	<0.001***
Licensing	4.3273	0.4357	9.933	<0.001***
Lic. : RC Neg.	-1.1481	0.4424	-2.59	<0.0095**
Unlic. : RC Neg.	1.4872	0.3372	4.411	**<0.001*** **

SENTENTIAL NEGATION IS AN ILLUSORY LICENSOR FOR BOTH NPIS AND NCIS AND THIS CAN'T BE EXPLAINED AS MISPARSING

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